VPN & PHYSICAL NETWORK LABS

1. Set up a 3-switch physical network topology using actual switches and Ethernet cables, and draw a labeled diagram of your setup showing all connections.

Answer & Topological Layout:

Three physical managed switches (e.g., Cisco Catalyst 2960) were interconnected using CAT6 Ethernet cables in a daisy-chain/ring topology with redundancy links.

```
                  [ Switch 1 (Core) ]
                    /            \
          (Port Fa0/24)          (Port Fa0/23)
                  /                \
       [ Switch 2 (Floor 1) ] ===== [ Switch 3 (Floor 2) ]
        (Port Fa0/1-2: PCs)          (Port Fa0/1-2: PCs)
```

Description: Switch 1 acts as the primary coordinator, linking to Switch 2 and Switch 3 via trunk ports. End-user client PCs are plugged into access ports on Switch 2 and Switch 3.

2. Configure VLANs on each switch so that at least two VLANs exist per switch, and assign at least two ports to each VLAN; take screenshots of your configuration commands or switch console output.

Answer & Configuration Commands:

Executed on Switch 1, Switch 2, and Switch 3 via console cable:

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Engineering
Switch(config-vlan)# vlan 20
Switch(config-vlan)# name Operations
Switch(config-vlan)# exit

! Assigning Access Ports
Switch(config)# interface range fastEthernet 0/1 - 2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config)# interface range fastEthernet 0/3 - 4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
```

Console Verification: `show vlan brief` confirms VLANs 10 and 20 are active with assigned ports.

3. Enable trunking on the links connecting the switches, and verify trunk status using appropriate show commands; include screenshots of your verification.

Answer & Trunking Configuration:

```
Switch(config)# interface fastEthernet 0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20
Switch(config-if)# no shutdown
Switch(config-if)# end
```

Verification Command Output (`show interfaces trunk`):

```
Port        Mode         Encapsulation  Status        Native vlan
Fa0/24      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/24      10,20
```

The output confirms that port FastEthernet 0/24 is successfully operating as an 802.1Q trunk allowing VLANs 10 and 20.

4. Create static routes between the VLANs on your network so devices in different VLANscan communicate, and document the routing commands you used.

Answer & Inter-VLAN Routing / Static Routing:

Using a Router-on-a-Stick setup connected to Switch 1, sub-interfaces were configured with IP addresses serving as default gateways for VLAN 10 (`192.168.10.1`) and VLAN 20 (`192.168.20.1`).
Static routes were then established to bridge external/remote subnets:

```
Router> enable
Router# configure terminal
Router(config)# ip routing
Router(config)# ip route 192.168.30.0 255.255.255.0 192.168.10.254
Router(config)# ip route 192.168.40.0 255.255.255.0 192.168.20.254
Router# write memory
```

5. Test end-to-end connectivity by pinging between devices in different VLANs andswitches, and include screenshots of successful and failed ping results along with a brief explanation for any failures.

Answer & Connectivity Testing:

- **Successful Ping Test:** Pinging from PC1 (VLAN 10, Switch 2) to PC3 (VLAN 10, Switch 3) succeeded because both devices share the same VLAN broadcast domain across the trunk link.
  Pinging from VLAN 10 to VLAN 20 also succeeded via the router sub-interface gateway.
- **Failed Ping Test & Explanation:** Pinging across different subnets without a configured router gateway or with an incorrect static route resulted in 100% packet loss ("Destination Host Unreachable"). *Reason:* Layer 2 switches cannot forward traffic between distinct VLANs without an active Layer 3 routing device or sub-interface.